IntelliDocX — AI-Driven Document Intelligence & Cognitive Processing Platform

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◈ 1. Project Overview

IntelliDocX is an AI-powered document intelligence platform that automates the extraction, understanding, classification, and management of unstructured documents.

The system integrates **Optical Character Recognition (OCR)** and **Large Language Models (LLMs)** with a **PostgreSQL-based relational architecture** to convert raw documents into structured, searchable, and actionable insights.

It is designed for ensuring scalability, security, and automation.

◈ 2. Objectives

- Automate document processing and reduce manual effort
- Extract structured data from unstructured files
- Enable intelligent document classification and routing
- Provide secure and role-based document access
- Support conversational interaction via AI chatbot

◈ 3. Key Features

Document Processing

- Upload PDF/image documents
- OCR-based text extraction
- Metadata generation

AI Intelligence

- LLM-based summarization
- Semantic classification
- Multilingual processing

Smart Routing

- Automatically assigns documents to departments

- Rule-based + AI-driven logic

Duplicate Detection

- Detects redundant documents
- Uses similarity scoring techniques

Security

- JWT authentication
- Role-Based Access Control (RBAC)
- Audit logging

Chatbot Integration

- AI chatbot for querying documents
- Natural language interaction

◈ 4. Technology Stack

Frontend

- React.js
- HTML5, CSS3, JavaScript
- Axios

Backend

- Node.js
- Express.js

Database

- PostgreSQL

AI & Processing

- OCR Engine (text extraction)
- LLM Adapter (summarization & classification)

Authentication

- JWT (JSON Web Tokens)

◈ 5. System Architecture

```
Frontend (React)
      ↓
REST API (Node.js + Express)
      ↓
Service Layer
  ├── OCR Module
  ├── LLM Processing Module
  ├── Routing Engine
  ├── Duplicate Detection Engine
      ↓
PostgreSQL Database
      ↓
Chatbot Microservice
```

◈ 6. Database Design

Users Table

- id (Primary Key)
- name
- email
- password_hash
- role (admin/user)
- created_at

Documents Table

- id (Primary Key)
- title
- file_path
- extracted_text
- summary
- department_id (Foreign Key)
- uploaded_by (Foreign Key → users)
- created_at

Departments Table

- id (Primary Key)

- name
- description

Audit Logs

- id
- user_id
- action
- timestamp

◈ 7. Workflow / Data Flow

1. User uploads a document
2. File is stored and processed by backend
3. OCR extracts text
4. LLM analyzes and summarizes content
5. Document is classified and routed
6. Duplicate detection is performed
7. Data is stored in PostgreSQL
8. User can search or interact via chatbot

◈ 8. API Modules

- **Auth API** → Login, Register
- **Document API** → Upload, Retrieve, Delete
- **Department API** → Manage categories
- **Routing API** → Assign documents
- **Integration API** → External services

◈ 9. Environment Variables

```
PORT=5000
DATABASE_URL=postgresql://user:password@localhost:5432/intellidocx
JWT_SECRET=your_secret_key
```

```
OCR_API_KEY=your_key
LLM_API_KEY=your_key
```

◈ 10. Installation & Setup

```
# Install dependencies
npm install

# Setup database (PostgreSQL)
npx prisma migrate dev
# or
npx sequelize db:migrate

# Start backend
npm start

# Start frontend
npm run dev
```

◈ 11. Security Considerations

- Password hashing (bcrypt)
- Token-based authentication (JWT)
- Role-based authorization
- Input validation & sanitization
- Secure API endpoints

◈ 12. Use Cases

- Enterprise document management
- Legal and compliance systems
- Government digital workflows
- Healthcare record processing
- Academic and research documentation

◈ 13. Future Enhancements

- Cloud deployment (AWS/GCP/Azure)
- Mobile application
- Advanced AI Capabilities
- Advanced Security
- Integration & Extensibility

◈ 14. Conclusion

IntelliDocX demonstrates how AI can transform traditional document workflows into intelligent, automated systems. By combining **OCR, LLMs, and PostgreSQL**, the platform ensures efficient, secure, and scalable document processing suitable for real-world enterprise environments.